LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

Quick Revision: Gravitation

Key terms

Term	Meaning / formula
Universal law of gravitation	$F = G\, m_1 m_2 / d^2$; $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}
Acceleration due to gravity (g)	$g = GM/R^2 \approx 9.8$ m/s^2 on Earth
Mass	Amount of matter; same everywhere; kg
Weight	Force of gravity on a body, W = mg; newton (N)
Thrust / pressure	Thrust is force acting perpendicular to a surface; pressure = thrust ÷ area, in pascal (Pa)
Buoyancy	The upward force a fluid exerts on an object in it

Must-know points

- If the distance between two bodies is doubled, the force becomes **one-fourth**.
- All objects fall with the same acceleration g in the absence of air resistance (free fall).
- Weight on the Moon is about **one-sixth** of the weight on Earth; mass does not change.
- g is slightly greater at the poles than at the equator, because the Earth's radius is smaller at the poles.
- Sharp knives cut easily and camels have broad feet: same force, smaller or larger area changes the pressure.
- **Archimedes' principle:** a body fully or partly immersed in a fluid feels an upward force equal to the weight of the fluid it displaces. An object floats if its density is less than the liquid's.

Equations of motion for free fall: replace a with g (and use −g for a body thrown up).
Relative density = density of the substance ÷ density of water (no unit)

Quick check

1. What happens to the gravitational force between two objects if the mass of one is doubled?

2. A 60 kg astronaut weighs how much on Earth and on the Moon? (g = 9.8 m/s^2)